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9

Improving Scan Insertion Run-Time and Capacity

Learning Objectives

After completing this lab, you should be able to:

- Given existing scan insertion scripts, implement ^③RSS in a top-down scan insertion flow achieving well-balanced scan chains
- Modify a bottom-up scan insertion script for full gate-level designs to use test models with RSS and run it
- Preview top-level chain balance using test models after block level scan insertion and revise the block level scan architecture as needed to improve top-level scan chain balance

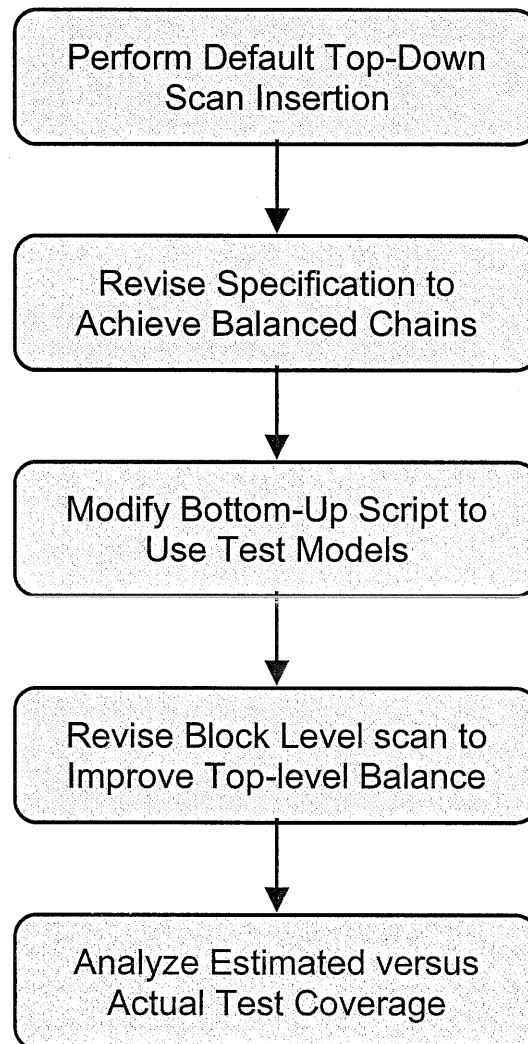


Lab Duration:
75 minutes

Lab 9

Overview

Lab 9 Tasks



Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question, check the back portion of this lab for help.

File Locations

All files for this lab are located in the directory **lab9_hicap**.

Directory Structure

lab9_hicap	Current working directory
logs	Session log files
mapped	ORCA design and its subblocks
mapped_scan	Scan-inserted design and subblocks
reports	DFT reports
scripts	Scan insertion scripts
test_models	Test models for subblocks
tmax	TetraMAX data
.solutions/scripts	Solutions for this lab

Relevant Files

scripts/	
bottom_up.tcl	Bottom-up scan insertion script. Calls the following two scripts:
scan_blocks.tcl	Inserts scan in to the subblocks
scan_top.tcl	Inserts scan in to the top-level
top_down.tcl	Top-Down scan insertion script

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Instructions

During this lab you will perform a top-down and a bottom-up scan insertion flow on the design ORCA. Some scripts are already provided. You will modify the scripts for the various tasks.

These scripts contain almost all the DFTC commands you would need to use when working on a real-world design back in your office. After completing this lab, you will have practiced the steps needed to improve both the run-time and the capacity of your scan insertion scripts. When you re-use the scripts back at your office, focus on the design-specific items (names of clocks, resets, and so forth).

The lab starts with a top-down script since it is easier to understand and provides reference data (number and length of top-level scan chains, estimated test coverage) that you can later use to compare the results of the bottom-up scan insertion script.

Task 1. Top-Down Scan Insertion

1. Make sure your current directory is **lab9_hicap**:

```
UNIX% cd lab9_hicap
```

2. Open the file `scripts/top_down.tcl` in your favorite editor (`vi`, `emacs`, etc.).
The `top_down.tcl` script is used to insert scan in a top-down fashion on the ORCA design.

```
UNIX% vi scripts/top_down.tcl
```

3. Modify this script to use the RSS features you learned about in lecture.

Compared to the original script in the lab directory, your revised script should:

-  a. Modify the original design as little as possible
- b. Run scan insertion faster
- c. Consume less memory

If needed, check the **answers** section for hints.

Once you are done, save the modified script file.

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4. Start Design Compiler from within the **lab9_hicap** directory:

```
UNIX% dc_shell | tee logs/lab9.log
```

5. Execute the script you just edited:

```
s scripts/top_down.tcl
```

6. Open the report generated by `preview_dft` and analyze it.

Question 1. Are the scan chains well balanced? If not, why?

.....

Note: It is important to balance the scan chains if tester time and the cost of test are to be reduced.

7. Open the **top_down.tcl** file again and **change** it so that you can achieve balanced top-level scan chains.
8. Remove the first pass of scan design files before improving the scan architecture.

```
remove_design -design
```

9. Rerun the `top_down.tcl` script in DFTC.

Question 2. Are the scan chains more balanced now?

.....

Question 3. What DFT structure did DFTC have to add to achieve these balanced scan chains?

.....

Note: The ORCA design has only 3 test clocks, but your real-world design might have many more test clocks. The more test clocks, the greater the probability for having more of these DFT structures. If minimizing die area is an important design requirement, you may need to monitor their number.

② How

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10. Determine the number of lockup latches DFTC inserted. DFTC names the latches using an intuitive naming convention that is good for wildcard searches.

```
get_cells LOCKUP* -hier
```

Question 4. How many lockup latches were added?

.....

Note: If your design has many lockup latches counting them yourself can become tedious. Have DFTC count them using:

```
sizeof_collection [get_cells LOCKUP* -hier]
```

Question 5. Where in the ORCA hierarchy were they placed?

.....

Note: If the ORCA design will be placed and routed using a top-down methodology, the placement of the lockup latches is probably not important. If the ORCA design will be placed and routed in a bottom-up fashion, however, you may have preferred locations for them. For example, you may want to avoid any lockup latches at the top-level.

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11. Obtain an estimate of the design's test coverage in DFTC.

Question 6. What is the estimated test coverage for ORCA?

.....

12. Exit Design Compiler.

Task 2. Bottom-Up Scan Insertion using Test Models

Two scripts are used for bottom-up scan insertion, `scan_blocks.tcl` and `scan_top.tcl`. These scripts currently run using full gate-level netlists.

In the subsequent steps you will first run `scan_blocks.tcl` to insert scan into the blocks of the ORCA design generating test models for the blocks. Later you will run the `scan_top.tcl` to complete the top-level scan chain stitching in ORCA using the test models created for the blocks instead of the full gate-level netlists.

The ORCA top-level design contains all the I/O pad instantiations, a `CLOCK_GEN` block that contains the PLLs and clock multipliers, and an `ORCA_TOP` block that serves as the core of the design and contains all the blocks for which you will perform scan insertion.

The `foreach` loop in the `scan_blocks.tcl` script iterates through a Tcl list of the blocks within `ORCA_TOP`. The loop first reads in the appropriate `.ddc` file from the correct directory and then proceeds with block-level scan insertion. The results are saved at the end of the loop and all design files are removed before proceeding to the next block.

1. Locate the `foreach` loop within the `scan_blocks.tcl` script and **modify it to save the test model for each block in a file separate from the DDC file in the `test_models` directory** (put these commands in the script immediately before the command that removes each block).
2. Modify the `scan_top.tcl` to **read in the test models only from the proper directory** (these test models were saved by the `scan_blocks.tcl` script).
Read the test models first, and then read in the block that contains them, `ORCA_TOP`.
3. Make sure the `RESET_BLOCK` is **not** a test model. **Read the `RESET_BLOCK` design from the “`mapped_scan`” directory.**

```

. . .
Your commands go here
. . .

read_ddc mapped/ORCA.ddc
current_design ORCA
link
. . .

```


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4. Examine the `scan_blocks.tcl` script and answer the following questions.

Question 7. Why specify `-clock_mixing no_mix` for block level scan insertion?

.....
.....

Question 8. What is the result of `-insert_terminal_lockup` for block level scan insertion?

.....
.....

Question 9. If lockup latches will be inserted at the end of all block level scan chains, do you need to insert any lockup latches between block level chains at the top-level?

.....

5. Run the `bottom_up.tcl` script, which first runs `scan_blocks.tcl` on the blocks within `ORCA_TOP` and then runs `scan_top.tcl` on the ORCA design.

```
UNIX% dc_shell -f scripts/bottom_up.tcl | tee  
bottom_up.log
```

6. Examine the top-level `preview_dft` report for ORCA.

```
vi reports/ORCA_preview_dft.rpt
```

Question 10. Are the scan chains balanced? If not, why?

.....

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7. Use the approach outlined in lecture to revise the block level scan architectures so the top-level scan chains are better balanced. Modify the `scan_blocks.tcl` script, rerun `bottom_up.tcl` and iterate as necessary to achieved better top-level balancing. Stop when all top-level scan chain lengths are within 20 scan cells of each other.

Question 11. Which command and setting produced this better-balanced top-level scan architecture?

.....

Question 12. In the context of a design that contains test models, why can you NOT use the same method for counting the number of lockup latches in the design as used in Task 1?

.....

.....

Note: Since the resulting Verilog gate-level netlists of both the block level scan insertion and the top-level scan stitching are saved to the same UNIX directory, you can use UNIX commands to count the lockup latches.

8. Determine the number of LOCKUP latches inserted throughout ORCA.

```
UNIX% fgrep LOCKUP tmax/*.v | wc -l
```

Note: `fgrep` searches for fixed string matches and `wc -l` outputs the number of matched lines. In this case, that number is equal to the number of lockup latches.

Question 13. How many lock-up latches were inserted?

.....

Question 14. Where in the ORCA hierarchy were they placed (*Hint: do the `fgrep` without piping it to `wc`*)?

.....

.....

.....

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Note: If the ORCA design will be placed and routed using a top-down methodology, the placement of the lockup latches is probably not important. If the ORCA design will be placed and routed in a bottom-up fashion, however, you may have preferred locations for them. For example you may want to avoid any lockup latches at the top-level.

Question 15. How does this compare to the number of lock-up latches inserted in Task 1 by the top-down script?

.....

9. Examine the estimated coverage report for ORCA and answer the following questions.

```
UNIX% cat reports/ORCA_coverage.rpt
```

Question 16. How does this coverage compare to the estimated coverage obtained when inserting scan top-down into ORCA?

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Question 17. Why is the estimated coverage obtained using test models not meaningful?

.....

Note: In order to obtain a more meaningful estimate for the test coverage, you must read the entire design into TetraMAX.

10. Run a script to obtain the actual test coverage in TetraMAX.

```
UNIX% cd tmax
UNIX% tmax -shell orca_tmax.tcl
```

Question 18. Why is the total number of faults in TetraMAX much greater than the number of faults in DFTC?

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Question 19. What test coverage does TetraMAX obtain for ORCA?

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Congratulations! You have completed this lab.